3. Methodology: Measuring Mechanistic Overlap

Step 1: Candidate Fact Pairs (N=100) We designed pairs spanning semantic categories: shared subject/object, shared tokens with multiple senses, and unrelated controls.

Step 2: Neuron-Level Overlap Scoring To measure polysemanticity, we compute a **FiNE-style contribution score** for neuron i at layer ℓ (using Pythia-160M probe):

$$c_i^\ell \;=\; \left|q_i^\ell \cdot \left(v_t^\top W_{\text{out}}^\ell\right)_i\right|$$

(Combines activation q_i^ℓ with projection to target token embedding v_t).

We identify Top-K ($K = 50$) neurons for Fact A (U_A) and Fact B (U_B) and measure **Jaccard Similarity**:

$$\text{Jaccard}(U_A, U_B) \;=\; \frac{|U_A \cap U_B|}{|U_A \cup U_B|}$$

4. Defining Experimental Regimes

We binned the 100 pairs based on Jaccard scores into mechanistic regimes:

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- **Set-Poly (High Overlap):** 50 pairs with highest scores (> 0.30).
- **Set-Mono (Low Overlap):** 50 pairs with lowest scores (< 0.15).

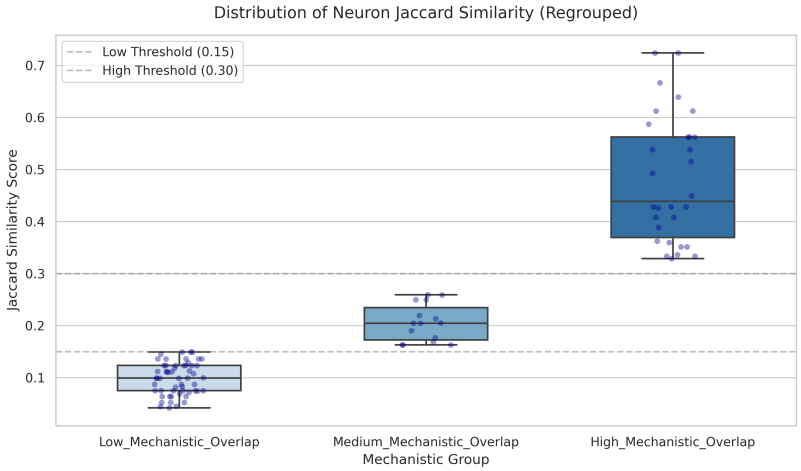


Figure 1: **Regrouped Distribution of Neuron Jaccard Similarity.** Shows partitioning into Low (Set-Mono), Medium, and High (Set-Poly) mechanistic overlap groups.